2 **(a)** Define gravitational potential at a point

The gravitational potential at a point is defined as the work done per unit mass [B1]

by an external force in bringing a small test mass from infinity to that point. [B1]

[2]

(b) **(i)** The dwarf planet Ceres may be considered to be an isolated uniform sphere of mass 9.4×10^{20} kg and radius 4.7×10^{5} m.

A particle is launched vertically upwards from the surface of the planet. Assuming negligible resistance from the planet's atmosphere, determine the minimum speed of this particle that will result in it escaping from the gravitational pull of the planet Ceres.

By conservation of energy,

Total GPE + KE at surface = Total GPE + KE at infinite distance away [C1]

$$\text{GPE}_{surface} + \text{KE}_{surface} = \text{GPE}_{r=\infty} + \text{KE}_{r=\infty}$$

$$-\frac{GMm}{r} + \frac{1}{2}mv^2 = 0 + 0$$

$$\frac{GM}{r} = \frac{1}{2}v^2$$

$$\frac{(6.67 \times 10^{-11})(9.4 \times 10^{20})}{4.7 \times 10^{5}} = \frac{1}{2}v^2 \quad \text{[C1]}$$

$v = 520 \text{ m s}^{-1}$ [A1]

speed = ... m s^{-1} [3]

(ii) Hydrogen may be assumed to be an ideal gas.
The mass of a hydrogen molecule is 3.3×10^{-27} kg.

Calculate the root-mean-square (r.m.s.) speed of a hydrogen molecule in hydrogen gas that is at a temperature of 240 K.

$$\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$$

$$\frac{1}{2}(3.3 \times 10^{-27})c_{rms}{}^2 = \frac{3}{2}(1.38 \times 10^{-23})(240) \quad \text{[C1]}$$

$$c_{rms} = 1700 \text{ m s}^{-1} \quad \text{[A1]}$$

r.m.s. speed = ... m s^{-1} [2]

(c) The surface of planet Ceres reaches temperatures of approximately 240 K when in direct sunlight.

Use your answers in **(b)(i)** and **(b)(ii)** to suggest a reason why Ceres does not have an atmosphere consisting of hydrogen.

Since r.m.s. speed of hydrogen molecules is more than the escape velocity on the surface of Ceres, the hydrogen molecules have sufficient kinetic energy to overcome the gravitational pull of the planet Ceres and escape to infinity. [1]

[Total: 8]